Q7 (refined). Why weren't the new transitions seen before?

Answer updated after reading Chand et al. 1991 directly

Correction to v1

My first pass leaned on Mattera's description of Chand as a *"limited two-detector setup."* Reading Chand directly, the characterization is fair but incomplete. Chand used **four** semiconductor detectors, reached sensitivity $\sim 3 \times 10^{-3}$ of the strongest line, and did real γ-γ coincidence work — just not the *kind* of coincidence work that finds new transitions. The refined story is more interesting than "they had fewer detectors."

What Chand et al. 1991 actually did

Their instrumentation (p. 92 of Chand):

- Horizontal planar Si(Li) (4–90 keV, 165 eV FWHM at 5.9 keV) — for L X-rays.
- Vertical planar HPGe (20–600 keV, 459 eV FWHM at 122 keV) — for K X-rays and low-E γ's.
- Two coaxial HPGe detectors (57.1 and 96.0 cm^3, 1.7 keV FWHM at 1332 keV) — for γ rays 80–4000 keV.
- Mini-orange electron spectrometer — for conversion electrons (their novel contribution).

The γ-γ coincidence setup used the two coaxial HPGe detectors (90 + 96 cm^3) with 7 ns timing. But **the purpose was angular correlation analysis of 15 specific cascades**, not a systematic search for new lines. They gated on the 131, 487, and 1596 keV transitions to measure $W(\theta)$ for known cascades and extract mixing ratios — identical physics to Fig. 9 in Mattera. They did not build a full 2D coincidence matrix and scan it for unknown peaks.

Energy resolution is comparable to Mattera's setup. Count rates were 1200 counts/s at 25 cm. Ten spectra per detector per source, 25,000–150,000 s each. This is a careful, well-executed singles measurement with targeted angular-correlation follow-up — not a crude experiment.

Transition-by-transition: why each new line was missed

Looking at Chand's Table 3, their sensitivity floor is around $I_\gamma \sim 3 \times 10^{-3}$ (e.g., the 445 keV line at 0.003). Mattera reaches 1.5×10^{-5} — two orders of magnitude deeper.

New transition	Intensity	vs. Chand floor	Why Chand missed it
166 keV	0.019	**above**	Buried between 131 (I=0.49) and 173 (I=0.13) in singles; sits on Compton c
645 keV	0.0023	at/below	Below singles sensitivity. Requires gating on the *weak* 307 keV line (itself I=
983 keV	0.009	near	In a crowded region (strong 993 keV transition right next door, I=0.007). C
1010 keV	0.008	near	Sits near the 1085 keV single-escape peak of the 1596 keV line (see Mattera

Key Concept

The 166 keV case is instructive. Its intensity (I=0.019) is **above Chand's nominal sensitivity floor**, so naïvely they should have seen it. They didn't, because "sensitivity floor" is local: a peak is visible only if its counts exceed the Poisson noise of the continuum under it. At 166 keV, the continuum from every stronger transition above 166 keV is piled up, plus it sits in the shoulder of the strong 131 and 173 keV peaks. Coincidence gating doesn't improve your *detector*; it removes the continuum itself.

The more interesting result: Chand got two transitions wrong

This is the part my first pass missed entirely. Chand reported two transitions that Mattera has now **removed** from the decay scheme:

Transition	Chand value	Mattera limit	Discrepancy
1097 keV (3001 → 1903)	$I_\gamma = 0.025(4)$	< 0.0002	125×
438.5 keV (2521 → 2083)	$I_\gamma = 0.041(10)$	< 0.005	8×

Both are disproved by coincidence gating in Mattera Figs. 5 and 6. Chand saw a peak at those energies in singles and assigned it to a specific cascade. In coincidence, gating on the expected partner transition (307 keV for 1097, 487 keV for 438.5) shows *no* peak where Chand claimed one.

From the paper

These are almost certainly **sum peaks or misidentifications** in Chand's singles spectrum, not real transitions. A sum peak at 1097 keV could arise from 307 + 790 or similar accidental combinations. Mattera's Table 1 does discuss this kind of issue for the 2083.7 keV transition, where the 487 + 1596 sum lands right on top of a real line. Chand had no way to correct for this cleanly without high-statistics coincidence data.

So modern coincidence gating does **two** jobs, not one: it finds weak transitions the old setup missed, AND it kills fake transitions the old setup invented. Both are improvements. The Mattera paper quietly does both.

The 2516 keV spin: I had this wrong

In my first pass I said "Chand assigned the 2516 keV level as $J^\pi = 3^+$, Mattera now says 4^+." Reading Chand directly (Sec. 4.3, p. 100):

From the paper

"It is clear from this comparison that it is not possible to distinguish clearly between the 3^+ and 4^+ spin assignment to the 2516 keV level ... The spin assignment to this level by the previous workers was done on the basis of weak arguments."

Chand was **agnostic**. They analyzed the 432 and 919 keV transitions under both hypotheses (their Table 8) and concluded the data couldn't distinguish. The 3^+ assignment came from Michelkakis and Hamilton 1982 (Chand's ref. 14); the 4^+ assignment came from Garcia Bermudez 1976 (ref. 13) and Kaur 1980 (ref. 9). Mattera's angular correlation (Fig. 9, Table II) now cleanly favors 4^+ by high-statistics correlation fits on multiple cascades.

So the story isn't "Chand was wrong and Mattera corrected them." It's *"Chand honestly said the data couldn't decide, and Mattera finally had enough statistics to decide."* That's a better and more interesting version of the narrative.

Refined bottom line

Bottom line

The new transitions weren't seen before because **Chand's methodology wasn't designed to find them**, not because their detectors were bad. They built a careful singles + targeted-coincidence experiment aimed at intensities, conversion coefficients, and mixing ratios for cascades they already knew about. That's different from a systematic coincidence-matrix scan for unknown transitions, which is what Gammasphere/CLARION2 enable.

The specific reasons per transition:

- **166 keV**: visible intensity, invisible location — buried in continuum.
- **645, 983, 1010 keV**: at or below Chand's statistical floor; require high-stats gating on weak partner transitions.

Two other results of the modern measurement that should probably be mentioned in your journal club because they're arguably more important than the new transitions:

- The 1097 keV and 438.5 keV transitions in Chand's decay scheme are **not real**. Coincidence gating kills fake peaks as well as finds real ones.
- The 2516 keV level spin is **settled at** 4^+ after 50 years of back-and-forth. Chand couldn't decide; Mattera could.

If anyone pushes you on "isn't two detectors enough for coincidences?": yes, technically, but the bottleneck isn't capability, it's statistics. Chand's coincidence setup was built to study ∼15 specific cascades with enough counts for angular fits. Gammasphere's 1128 detector pairs collect coincidence data ∼1000× faster, enabling a different kind of analysis entirely — systematic discovery rather than targeted follow-up.

One thing worth flagging in your journal club

The Chand paper is actually quite self-aware about its limits. They explicitly discuss which transitions they confirmed from prior work (Kaur 1980's 1303 keV, Adam 1982's 992 and 1877 keV) and where their data agrees/disagrees with prior measurements. They treat the 2516 keV spin as genuinely open. This is exactly the kind of careful 1991-era measurement that *should* be the foun-

dation of the ENSDF evaluation — it's just that the sensitivity and statistics of the era couldn't reach the 10^{-5} regime or systematically clean up cross-over transition placements.

So the Mattera paper isn't "correcting bad old work." It's saying: *the best 1991 measurement was this careful Panjab paper, we have better instruments now, here's what we can add.* That's a healthier framing and probably worth echoing in your presentation.